

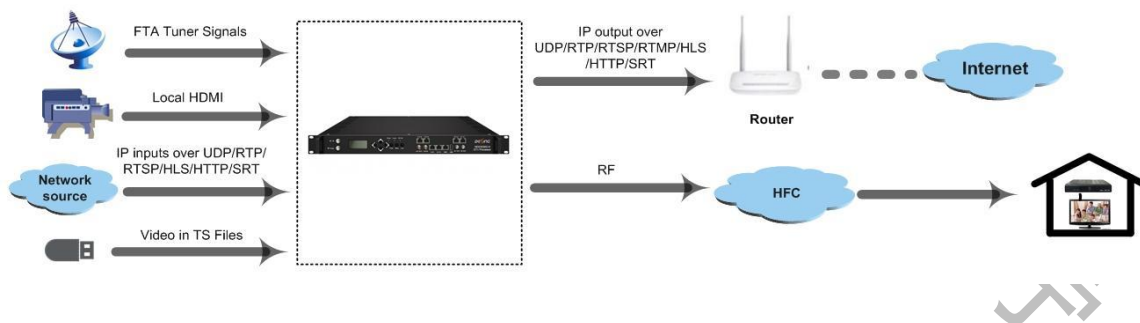
FMUSER FUQ-4647B-N DTV Processor 8-24 HDMI 1 ASI 1 USB 128 IP Inputs to 32 DVB-C 16 DVB-T-ATSC-ISDB-T Outputs HDMI-ASI-USB-IP Interfaces HEVC-H.265 AVC-H.264 Encoding Integrated DVB IPTV Streaming for Terrestrial Cable IPTV Headends

Product Overview



The FMUSER FUQ-4647B-P DTV Processor integrates DVB encoding and IPTV streaming in one chassis. It supports inputs including HDMI, ASI, USB, and IP, and outputs for RF and IP streams. With web-based control, it is ideal for broadcast applications in DVB and IPTV environments.

System Application



Key feature

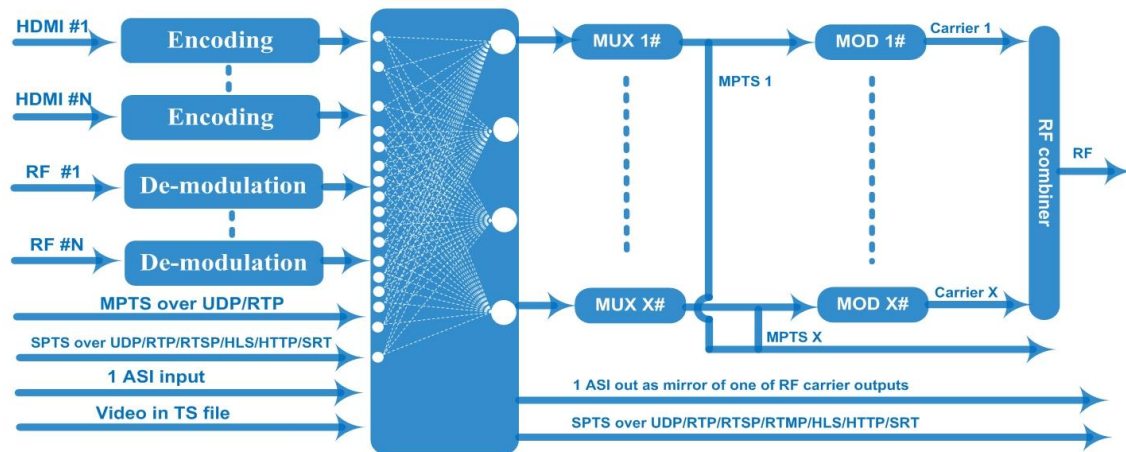
Integrated DVB encoding and IPTV streaming in a single unit.

Supports up to 24 HDMI inputs, 1 ASI input, 1 USB player, and 128 IP inputs.

Delivers up to 32-channel RF output for DVB-C/T/ATSC-ISDB-T and IP streaming.

Features HEVC/H.265 and MPEG-4 AVC/H.264 encoding with web-based management.

Working Block



Technical Specifications

Input	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 MPTS IP input over UDP and RTP, GE port, RJ45 SPTS input over UDP/ RTP/RTSP/ HLS/HTTP/ SRT	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 MPTS IP input over UDP and RTP, GE port, RJ45 SPTS input over UDP/ RTP/RTSP/ HLS/HTTP/ SRT	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 MPTS IP input over UDP and RTP, GE port, RJ45 SPTS input over UDP/ RTP/RTSP/ HLS/HTTP/ SRT	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 MPTS IP input over UDP and RTP, GE port, RJ45 SPTS input over UDP/ RTP/RTSP/ HLS/HTTP/ SRT	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 MPTS IP input over UDP and RTP, GE port, RJ45 SPTS input over UDP/ RTP/RTSP/ HLS/HTTP/ SRT	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 MPTS IP input over UDP and RTP, GE port, RJ45 SPTS input over UDP/ RTP/RTSP/ HLS/HTTP/ SRT
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Video (HDMI Input)	Resolution	Input	Input	Input	1920×1080_60P, 1920×1080_60i, 1920×1080_50P, 1920×1080_50i, 1280×720_60P, 1280×720_50P, 720×576_50i, 720×480_60i,	1920×1080_60P, 1920×1080_60i, 1920×1080_50P, 1920×1080_50i, 1280×720_60P, 1280×720_50P, 720×576_50i, 720×480_60i,
Video (HDMI Input)	Resolution	Output	Output	Output	1920×1080_30P, 1920×1080_25P, 1280×720_30P, 1280×720_25P, 720×576_25P, 720×480_30P,	1920×1080_30P, 1920×1080_25P, 1280×720_30P, 1280×720_25P, 720×576_25P, 720×480_30P,
Video (HDMI Input)	Encoding	HEVC/H.265 & MPEG-4 AVC/H.264	HEVC/H.265 & MPEG-4 AVC/H.264	HEVC/H.265 & MPEG-4 AVC/H.264	HEVC/H.265 & MPEG-4 AVC/H.264	HEVC/H.265 & MPEG-4 AVC/H.264
Video (HDMI Input)	Bit-rate	1Mbps~13Mbps each channel	1Mbps~13Mbps each channel	1Mbps~13Mbps each channel	1Mbps~13Mbps each channel	1Mbps~13Mbps each channel
Video (HDMI Input)	Rate Control	CBR/VBR	CBR/VBR	CBR/VBR	CBR/VBR	CBR/VBR
Video (HDMI Input)	GOP Structure	IP...P (P Frame adjustment)	IP...P (P Frame adjustment)	IP...P (P Frame adjustment)	IP...P (P Frame adjustment)	IP...P (P Frame adjustment)
Audio (HDMI Input)	Encoding	MPEG-1 Layer 2, LC-AAC, HE-AAC and AC3 Pass through	MPEG-1 Layer 2, LC-AAC, HE-AAC and AC3 Pass through	MPEG-1 Layer 2, LC-AAC, HE-AAC and AC3 Pass through	MPEG-1 Layer 2, LC-AAC, HE-AAC and AC3 Pass through	MPEG-1 Layer 2, LC-AAC, HE-AAC and AC3 Pass through
Audio (HDMI Input)	Sampling rate	48KHz	48KHz	48KHz	48KHz	48KHz

Audio (HDMI Input)	Resolution	24-bit	24-bit	24-bit	24-bit	24-bit
Audio (HDMI Input)	Audio Gain	0-255 Adjustable	0-255 Adjustable	0-255 Adjustable	0-255 Adjustable	0-255 Adjustable
Audio (HDMI Input)	MPEG-1 Layer 2 Bit-rate	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps
Audio (HDMI Input)	LC-AAC Bit-rate	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps
Audio (HDMI Input)	HE-AAC Bit-rate	48/56/64/80/96/112/128 kbps	48/56/64/80/96/112/128 kbps	48/56/64/80/96/112/128 kbps	48/56/64/80/96/112/128 kbps	48/56/64/80/96/112/128 kbps
Tuner Section	DVB-S	Input Frequency	Input Frequency	950-2150M Hz	950-2150M Hz	950-2150M Hz
Tuner Section	DVB-S	Symbol Rate	Symbol Rate	0.5~45 Msps	0.5~45 Msps	0.5~45 Msps
Tuner Section	DVB-S	Constellation	Constellation	QPSK	QPSK	QPSK
Tuner Section	DVB-S	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Tuner Section	DVB-S2	Input Frequency	Input Frequency	950-2150M Hz	950-2150M Hz	950-2150M Hz
Tuner Section	DVB-S2	Symbol Rate	Symbol Rate	QPSK/8PSK/16APSK: 0.5~45 Msps 32APSK: 0.5~40 Msps	QPSK/8PSK/16APSK: 0.5~45 Msps 32APSK: 0.5~40 Msps	QPSK/8PSK/16APSK: 0.5~45 Msps 32APSK: 0.5~40 Msps
Tuner Section	DVB-S2	Constellation	Constellation	QPSK/8PSK/16APSK/32APSK	QPSK/8PSK/16APSK/32APSK	QPSK/8PSK/16APSK/32APSK

Tuner Section	DVB-S2	FEC	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9
Tuner Section	DVB-S2	Max input bitrate	Max input bitrate	≤170 Mbps per tuner	≤170 Mbps per tuner	≤170 Mbps per tuner
Tuner Section	DVB-S2X	Symbol Rate	Symbol Rate	QPSK/8PSK/16APSK: 0.5~45 Msps; 8APSK/32APSK: 0.5~40 Msps	QPSK/8PSK/16APSK: 0.5~45 Msps; 8APSK/32APSK: 0.5~40 Msps	QPSK/8PSK/16APSK: 0.5~45 Msps; 8APSK/32APSK: 0.5~40 Msps
Tuner Section	DVB-S2X	Constellation	Constellation	QPSK/8PSK/16APSK/8APSK/32APSK	QPSK/8PSK/16APSK/8APSK/32APSK	QPSK/8PSK/16APSK/8APSK/32APSK

Tuner Section	DVB-S2X	FEC	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9
Tuner Section	DVB-S2X	Max input bitrate per tuner	Max input bitrate per tuner	≤170 Mbps	≤170 Mbps	≤170 Mbps
Tuner Section	Signal Strength	-65 ~ -25d Bm	-65 ~ -25d Bm	-65 ~ -25d Bm	-65 ~ -25d Bm	-65 ~ -25d Bm
Multiplexing	Maximum PID Remapping	255 input per channel	255 input per channel	255 input per channel	255 input per channel	255 input per channel

Multiplexing	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically
Modulation	DVB-C	Standard	Standard	Standard	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B
Modulation	DVB-C	RF output channel	RF output channel	RF output channel	32 non-adjacent carriers output (maximum bandwidth 384MHz)	32 non-adjacent carriers output (maximum bandwidth 384MHz)
Modulation	DVB-C	MER	MER	MER	≥40db	≥40db
Modulation	DVB-C	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-C	RF output level	RF output level	RF output level	-20~+1dbm, 0.1db step	-20~+1dbm, 0.1db step
Modulation	DVB-C	Symbol Rate	Symbol Rate	Symbol Rate	5.0Msps~7.0Msps, 1ksps stepping	5.0Msps~7.0Msps, 1ksps stepping
Modulation	DVB-C				J.83A	J.83B
Modulation	DVB-C	Constellation	Constellation	Constellation	16/32/64/128/256QAM	64/256 QAM
Modulation	DVB-C	Bandwidth	Bandwidth	Bandwidth	8M	6M
Modulation	DVB-T	Standard	Standard	Standard	EN300744	EN300744
Modulation	DVB-T	FFT mode	FFT mode	FFT mode	2K,4K, 8K	2K,4K, 8K

Modulation	DVB-T	Bandwidth	Bandwidth	Bandwidth	6M, 7M, 8M	6M, 7M, 8M
Modulation	DVB-T	Constellation	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	DVB-T	Guard Interval	Guard Interval	Guard Interval	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32
Modulation	DVB-T	Code Rate	Code Rate	Code Rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	DVB-T	MER	MER	MER	≥42 dB	≥42 dB
Modulation	DVB-T	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-T	RF output channel	RF output channel	RF output channel	16 non-adjacent carriers output (maximum bandwidth 192MHz)	16 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	DVB-T	RF output level	RF output level	RF output level	-20~+1dBm, 0.1db step	-20~+1dBm, 0.1db step
Modulation	ATSC	Standard	Standard	Standard	ATSC A/53	ATSC A/53
Modulation	ATSC	Bandwidth	Bandwidth	Bandwidth	6M	6M
Modulation	ATSC	Constellation	Constellation	Constellation	8VSB	8VSB
Modulation	ATSC	FEC	FEC	FEC	RS(208 18 8)+Trellis	RS(208 18 8)+Trellis
Modulation	ATSC	MER	MER	MER	≥40dB	≥40dB
Modulation	ATSC	ACL	ACL	ACL	-55 dBc	-55 dBc
Modulation	ATSC	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step

Modulation	ATSC	RF output channel	RF output channel	RF output channel	16 non-adjacent carriers output (maximum bandwidth 192MHz)	16 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	ATSC	RF output level	RF output level	RF output level	-20~+3dbm (for all carriers), 0.5db stepping	-20~+3dbm (for all carriers), 0.5db stepping
Modulation	ISDB-T	Standard	Standard	Standard	ARIB STD-B31	ARIB STD-B31
Modulation	ISDB-T	Bandwidth	Bandwidth	Bandwidth	6M	6M
Modulation	ISDB-T	Constellation	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	ISDB-T	Guard Interval	Guard Interval	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Modulation	ISDB-T	FFT Mode	FFT Mode	FFT Mode	2K, 4K, 8K	2K, 4K, 8K
Modulation	ISDB-T	Code rate	Code rate	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	ISDB-T	MER	MER	MER	≥40dB	≥40dB
Modulation	ISDB-T	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	ISDB-T	RF output channel	RF output channel	RF output channel	16 non-adjacent carriers output (maximum bandwidth 192MHz)	16 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	ISDB-T	RF output level	RF output level	RF output level	-20dBm~+1dBm, 0.1dB stepping	-20dBm~+1dBm, 0.1dB stepping

MPTS output	1 ASI output as mirror of one of RF output carriers 32 MPTS output over UDP and RTP/RTSP as mirror of 32 DVB-C carriers(or 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-T/ ATSC/ISD B-T carriers for option), 1*1000M Base-T Ethernet interface, GE port	1 ASI output as mirror of one of RF output carriers 32 MPTS output over UDP and RTP/RTSP as mirror of 32 DVB-C carriers(or 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-T/ ATSC/ISD B-T carriers for option), 1*1000M Base-T Ethernet interface, GE port	1 ASI output as mirror of one of RF output carriers 32 MPTS output over UDP and RTP/RTSP as mirror of 32 DVB-C carriers(or 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-T/ ATSC/ISD B-T carriers for option), 1*1000M Base-T Ethernet interface, GE port	1 ASI output as mirror of one of RF output carriers 32 MPTS output over UDP and RTP/RTSP as mirror of 32 DVB-C carriers(or 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-T/ ATSC/ISD B-T carriers for option), 1*1000M Base-T Ethernet interface, GE port	1 ASI output as mirror of one of RF output carriers 32 MPTS output over UDP and RTP/RTSP as mirror of 32 DVB-C carriers(or 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-T/ ATSC/ISD B-T carriers for option), 1*1000M Base-T Ethernet interface, GE port	1 ASI output as mirror of one of RF output carriers 32 MPTS output over UDP and RTP/RTSP as mirror of 32 DVB-C carriers(or 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-T/ ATSC/ISD B-T carriers for option), 1*1000M Base-T Ethernet interface, GE port
SPTS output	SPTS over UDP/RTP/ RTSP/RTM P/HLS/HTT P/SRT	SPTS over UDP/RTP/ RTSP/RTM P/HLS/HTT P/SRT	SPTS over UDP/RTP/ RTSP/RTM P/HLS/HTT P/SRT	SPTS over UDP/RTP/ RTSP/RTM P/HLS/HTT P/SRT	SPTS over UDP/RTP/ RTSP/RTM P/HLS/HTT P/SRT	SPTS over UDP/RTP/ RTSP/RTM P/HLS/HTT P/SRT
System function	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)
System function	English language	English language	English language	English language	English language	English language
System function	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
Miscellaneous	Dimension (W×L×H)	Dimension (W×L×H)	482mm×328mm×44mm	482mm×328mm×44mm	482mm×328mm×44mm	482mm×328mm×44mm

Miscellaneous	Environment	Environment	0~45°C (work) -20~80°C Storage ■	0~45°C (work) -20~80°C Storage ■	0~45°C (work) -20~80°C Storage ■	0~45°C (work) -20~80°C Storage ■
Miscellaneous	Power requirements	Power requirements	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz

Ordering Guide

Input Section:

	HDMI	DVB-S2/S2X	DVB-C/T/T2/ISDB-T
Option 1	8	/	/
Option 2	16	/	/
Option 3	24	/	/
Option 4	8	8	/
Option 5	8	16	/
Option 6	16	8	/
Option 7	8	/	8
Option 8	8	/	16
Option 9	16	/	8
Option 10	8	8	8
Option 11	16	4	4

Output Section:

RF output standard	RF carrier channels
DVB-C out	32
DVB-T out	16
ATSC out	16
ISDB-T out	16

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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